INTEGRAL UNIVERSITY

Integral Business School

TOPIC

Retail Sales & Customer Behaviour Analytics

Business Intelligence

BM277



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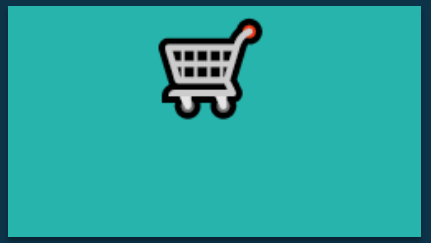
SUBMITTED TO

Mr Rahul Bisht

Retail Sales & Customer Behaviour Analytics

SQL + Power BI | Semester Project

Better insights. Smarter decisions.



50,000+ records

Jan 2023 – Dec 2024

Introduction

Turning retail transactions into customer and business insights.

About the Project

- Analyse retail sales data and understand customer behaviour.
- Use SQL for cleaning, querying, joins, aggregations and KPIs.
- Use Power BI to turn findings into interactive visuals.
- Identify product, customer and regional patterns.
- Support better customer experience and data-driven decisions.

1

Analyse

sales trends

2

Understand

customer behaviour

3

Identify

products & regions

4

Recommend

actions

Dataset Overview

The analysis combines sales, customer, product and store dimensions.

Data Sources

- Sales data
- Customer data
- Product data
- Store / region data

Key Tables

Sales • Customers
Products • Stores

Coverage

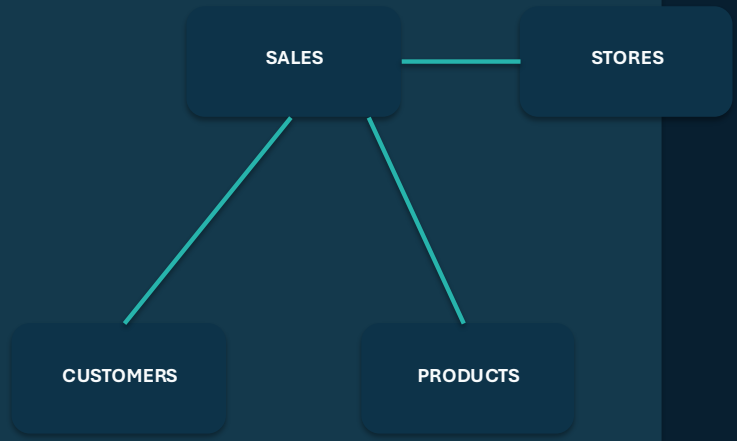
50,000+

total records

Jan 2023 – Dec 2024

24 months of retail activity

Data Model



SQL Implementation

Example queries used to answer core business questions.

01 Total sales by product category

```
SELECT p.category, SUM(s.sales_amount) AS total_sales
FROM sales s JOIN products p ON s.product_id = p.product_id
GROUP BY p.category ORDER BY total_sales DESC;
```

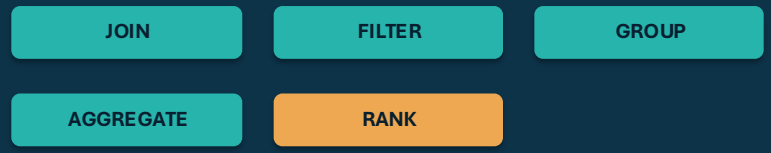
02 Monthly sales trend

```
SELECT DATE_TRUNC('month', order_date) AS month,
SUM(sales_amount) AS total_sales
FROM sales GROUP BY month ORDER BY month;
```

03 Top customers by total spend

```
SELECT c.customer_id, c.name, SUM(s.sales_amount) AS total_spend
FROM sales s JOIN customers c ON s.customer_id = c.customer_id
GROUP BY c.customer_id, c.name ORDER BY total_spend DESC LIMIT 5;
```

SQL workflow



Power BI Dashboard Overview

Executive snapshot of the retail business.

TOTAL SALES

$4,892,671

+12.5% vs last year

TOTAL ORDERS

48,320

+10.8% vs last year

AVG. ORDER VALUE

$101.23

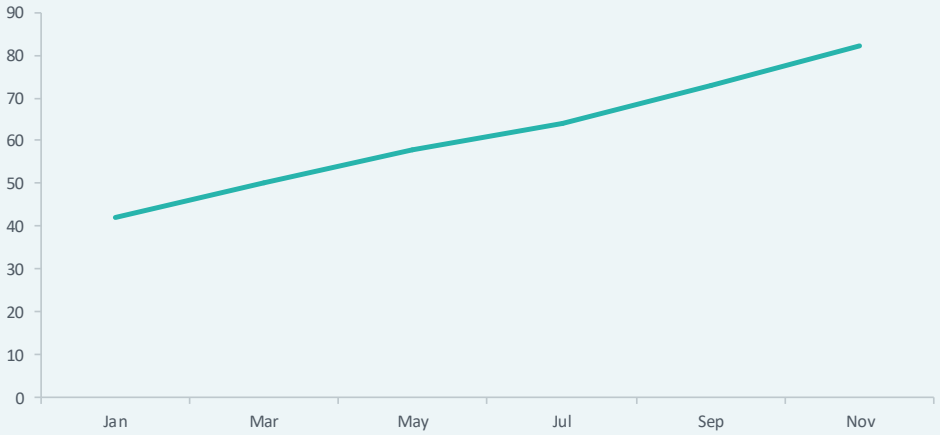
+8.7% vs last year

TOTAL CUSTOMERS

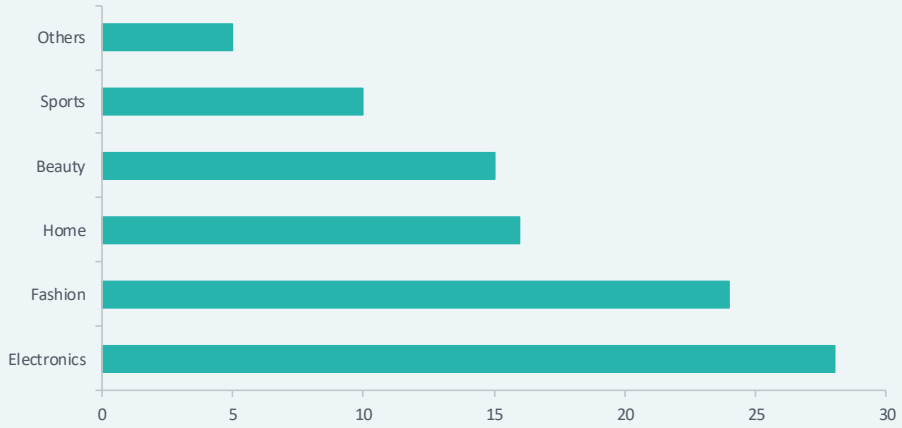
12,846

+14.2% vs last year

Sales trend



Sales by category



Key Insights

Five findings that matter most to the business.

01

Sales Trend

Steady growth with a 12.5% increase YoY.
Peak sales occur in Nov–Dec (festive season).

02

Best Category

Electronics contributes 28% of total sales.
Fashion and Home & Living are also strong.

03

Customer Behaviour

18–34 age group drives 62% of customers.
Male customers contribute 54% of revenue.

04

Regional Performance

North region leads with 28% of sales.
South and West show high growth potential.

05

Top Products

Laptops, smartphones and headphones are the leading products by sales.

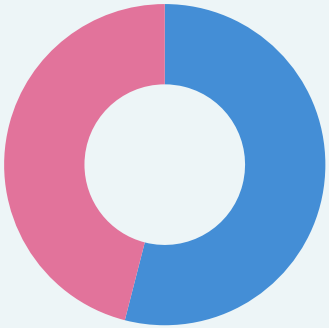
Customer Segmentation

Who buys, how valuable they are, and how they behave.

Customers by age



Customers by gender



RFM analysis

High value	18%
Medium value	47%
Low value	35%

Customer segments

Premium

High spenders

Regular

Moderate spenders

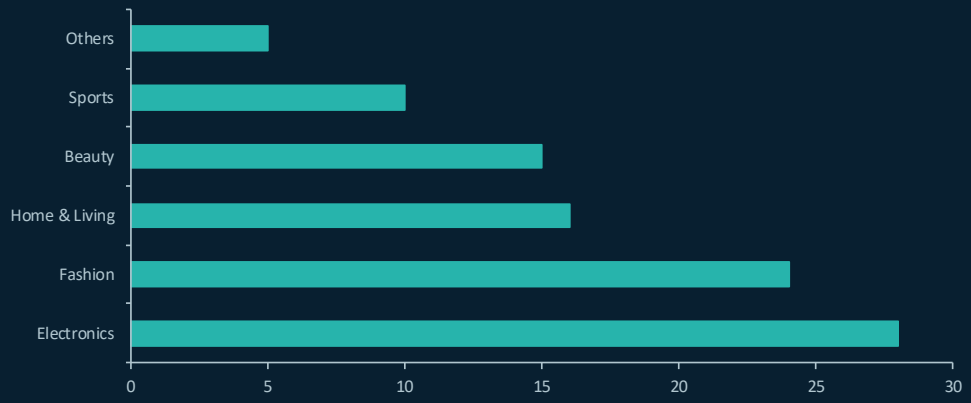
Occasional

Low spenders

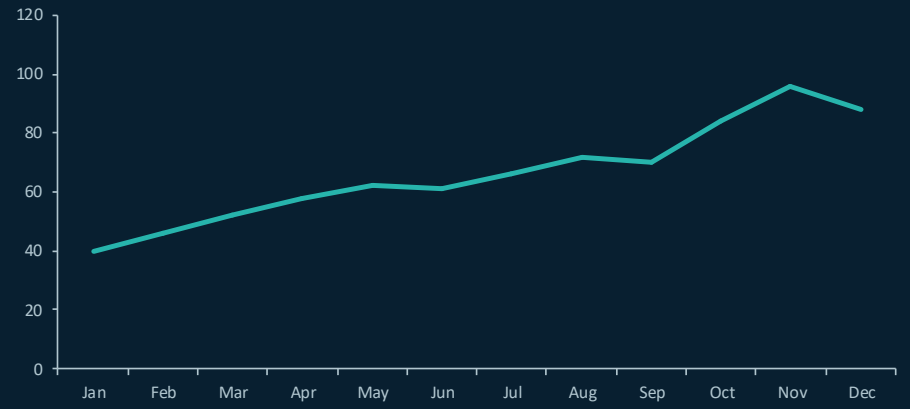
Sales & Category Analysis

Category mix, regional contribution and key takeaways.

Sales by category



Monthly sales trend



KEY TAKEAWAYS

- ✓ Electronics is the top revenue generator.
- ✓ Sales peak during festive months.
- ✓ Fashion shows consistent growth.

Customer Purchase Behaviour

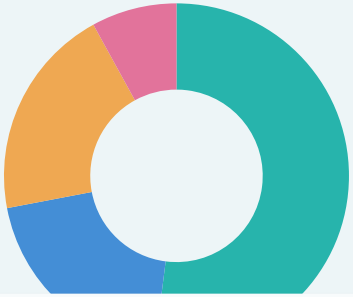
How customers pay and whether they return.

AVERAGE ORDER VALUE

$101.23

+8.7% vs last year

Preferred payment method



Repeat vs new customers



Key behavioural insights

- 42% of customers are repeat buyers.
- UPI is the most preferred payment method (52%).
- 18–34 age group has the highest AOV.

52% customers prefer UPI

SQL Code Snippets

Queries for customer ranking and inventory-style analysis.

03 Top 5 customers by total spend

```
SELECT c.customer_id, c.name,
    SUM(s.sales_amount) AS total_spend
FROM sales s
JOIN customers c ON s.customer_id = c.customer_id
GROUP BY c.customer_id, c.name
ORDER BY total_spend DESC
LIMIT 5;
```

Purpose Ranks customers by lifetime / period spend.

04 Products with low stock

```
SELECT product_id, name, stock_quantity
FROM products
WHERE stock_quantity < 10
ORDER BY stock_quantity;
```

Purpose Flags products that may need replenishment.

SQL → INSIGHT

QUERY → DECISION

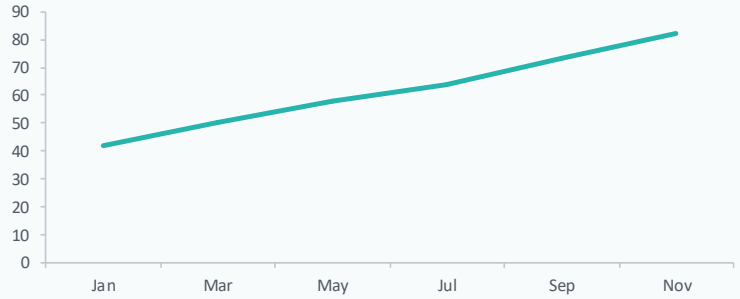
Power BI Visuals

Sample executive dashboard layout.

Retail Sales Dashboard

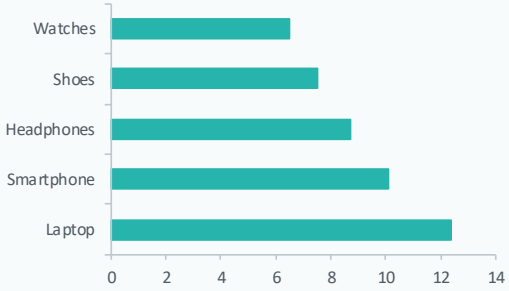
Total Sales	Orders	AOV	Customers
$4.89M	48.3K	$101.23	12.8K

Sales trend

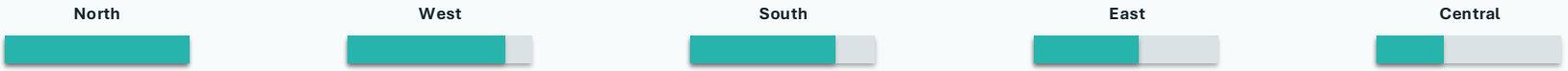




Top products



Region performance



Findings & Recommendations

From descriptive analytics to actions the business can take.

Key Findings

- Electronics and Fashion are the top categories.
- Festive season drives maximum sales.
- 18–34 is the largest customer group.
- UPI is the most used payment method.
- North region has the highest sales.

Recommendations

01 **Increase stock & promotion**
for top-performing categories.

02 **Launch targeted campaigns**
for 18–34 age group.

03 **Offer discounts**
on UPI payments.

04 **Explore expansion**
in South & East regions.

05 **Introduce loyalty programs**
to boost repeat purchases.

Business Impact

What better analytics enables.

1

Higher Revenue

Focus on high-performing categories and regions.

2

Better Customer Experience

Personalized offers and loyalty programs.

3

Operational Efficiency

Optimize inventory and reduce low-stock issues.

4

Data-Driven Decision Making

Use insights for strategic growth and planning.

Tools & Technologies

The project connects data preparation, analysis, visualization and collaboration.

SQL

Data cleaning, querying, joins, aggregations, KPIs

Power BI

Interactive dashboards, visualisation, insights

Excel

Data preparation and validation

GitHub

Version control and repository

VS Code

SQL editing and project workflow

“The best decisions are data-driven.”

Turning Data into Better Shopping Experiences

Through SQL and Power BI, the project turns retail data into clear trends, customer insights and practical recommendations.

4,892,671

total sales analysed

12,846

customers understood

THANK YOU!

Questions & Discussion